

5

The question of invariance

This chapter has a carefully chosen title: we are concerned here primarily with the question, and not with the answer. The question is this: suppose that K and K_1 , are simplicial complexes triangulating the same object, i.e. with $|K| = |K_1|$. Is it true that $H_p(K) \cong H_p(K_1)$ for all p ? The answer is in fact ‘yes’ – indeed a stronger assertion (5.13) is true, namely that the isomorphisms hold if $|K|$ is merely supposed homeomorphic to $|K_1|$, i.e. if there exists a continuous and bijective map $f: |K| \rightarrow |K_1|$. (The inverse map f^{-1} will automatically be continuous by a theorem of general topology, using the ‘compactness’ of $|K|$. See the books of Kelley or Lawson listed in the References.) The stronger assertion is referred to as the topological invariance of homology groups: it says that homology groups do not depend on particular triangulations but only on the underlying topological structure of $|K|$.

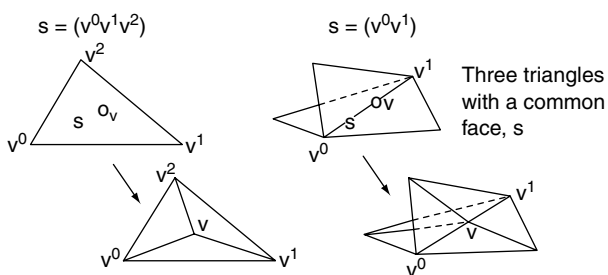
The topological invariance result can be restated in a negative form: if there exists an integer p such that $H_p(K)$ and $H_p(K_1)$ are not isomorphic, then $|K|$ and $|K_1|$ are not homeomorphic. Using results of Chapter 6 this in turn implies that no two inequivalent closed surfaces (2.8) have homeomorphic underlying spaces.

But we shall not prove the topological invariance theorem 5.13, since it would divert attention from the developments of succeeding chapters. The only invariance result actually needed is a much weaker one, and that one is proved in full; see 5.7.

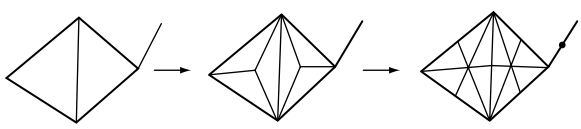
Though this chapter up to 5.9 forms an integral part of the logical structure of the book, it could be omitted on a first reading. It might be a good idea to read the first part of the section ‘Triangulations, simplicial approximation and topological invariance’, say up to 5.10, which expands on the general statements above.

Invariance under stellar subdivision

Let K be a simplicial complex. A *subdivision* of K is a simplicial complex K_1 with the property that $|K_1| = |K|$ and given $s_1 \in K_1$ there exists $s \in K$ such that $s_1 \subset s$. Thus the simplexes of K_1 are contained in simplexes of K but K_1 and K triangulate the same subset of $\mathbb{R}^N$. An example is provided by the barycentric subdivision of a surface described following 2.6, where each triangle is divided into six by drawing the medians, and each edge is thereby halved; more of this anon. Another example is the stellar subdivision described in 3.29(5): a reader who does not want to work through that example should read what follows by specializing always to the case where K has dimension 2. That will suffice for the applications in this book and will give a perfectly adequate idea of the techniques involved. A *stellar subdivision* is the result of a sequence of subdivisions, each of these being *starring* of a simplex s at an interior point v . When K has dimension 2, the following pictures show the result of starring s at an interior point v . When s is a vertex no change takes place. In general, we remove $\text{St}(s, K)$ and replace it by the join $v \star \text{Lk}(s, K)$; compare 3.24, 3.26.



When K has dimension 2 we can produce the first barycentric subdivision of K by starring each 2-simplex at its centroid and then starring each 1-simplex of the original simplicial complex at its mid-point.



In fact the procedure of starring by descending dimension always produces the first barycentric subdivision; a general definition of barycentric subdivision is given in an appendix to this chapter for those who wish to verify this fact.

Let K_1 be obtained from K by starring $s = (v^0 \dots v^n)$ at v . Then all simplexes of K with s as a face are deleted, and certain simplexes, listed below, are introduced. If K is oriented, then K_1 can be oriented in a natural way, and the orientations are indicated in the list, where it is assumed that $(v^0 \dots v^n)$ gives the correct orientation for s .

- (v), often written just v
 $(v^0 \dots v \dots v^n)$ where v replaces v^i in s ($i = 0, \dots, n$)
5.1 $\left. \begin{array}{l} \pm(vw^0 \dots w^m) \\ \pm(v^0 \dots v \dots v^n w^0 \dots w^m) \\ v \text{ replacing } v^i \text{ in } s \text{ (} i = 0, \dots, n \text{)} \end{array} \right\} \begin{array}{l} \text{where } \pm(v^0 \dots v^n w^0 \dots w^m) \\ \text{(same sign as on the} \\ \text{left) belongs to } K \end{array}$

The reader should check in some simple cases that these give the ‘right’ orientations for simplexes of K_1 . Note that

$$(v^0 \dots v \dots v^n) = (-1)^i (vv^0 \dots \widehat{v^i} \dots v^n)$$

where v replaces v^i in s on the left of the equation.

Some notation will be useful in what follows. Write σ for the oriented simplex $(v^0 \dots v^n)$, σ^i for $(v^0 \dots \widehat{v^i} \dots v^n)$, τ for $(w^0 \dots w^m)$ and τ^j for $(w^0 \dots \widehat{w^j} \dots w^m)$. Joins are denoted as usual by concatenation of symbols ($\sigma\tau$ etc.) and $v\sigma$, for example, denotes $(v\sigma) = (vv^0 \dots v^n)$. The notation can be extended to chains: for example $(\partial\sigma)\tau$ means $\Sigma(-1)^i(\sigma^i\tau) \in C_{m+n}(K)$.

Note that, in this notation, we have the following formula for the boundary of a join:

$$\mathbf{5.2} \quad \partial(\sigma\tau) = (\partial\sigma)\tau - (-1)^n\sigma(\partial\tau), \quad n = \dim \sigma.$$

(If $\dim \sigma = 0$ then $(\partial\sigma)\tau$ is to be read as τ , and similarly if $\dim \tau = 0$ then $\sigma(\partial\tau) = \sigma$.)

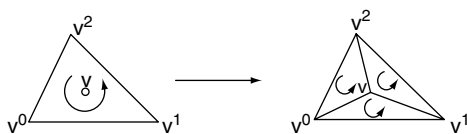
Now we can define a homomorphism

$$\alpha_p: C_p(K) \rightarrow C_p(K_1)$$

as follows. Let $\sigma\tau = (v^0 \dots v^n w^0 \dots w^m)$ be a simplex in $\text{St}(s, K)$; then

$$\alpha_p(\sigma\tau) = v(\partial\sigma)\tau = \sum_{i=0}^n (v^0 \dots v \dots v^n w^0 \dots w^m)$$

where v replaces v^i . This replaces $\sigma\tau$ by the sum of simplexes into which it is broken, with appropriate orientations. Also define $\alpha_p\sigma = v(\partial\sigma) = \Sigma(v^0 \dots v \dots v^n)$ (see diagram).



Other simplexes, not in $\text{St}(s, K)$, α_p leaves unchanged.

Proposition 5.3. α_p takes cycles to cycles and boundaries to boundaries and so induces a homomorphism

$$\alpha_* : H_p(K) \rightarrow H_p(K_1)$$

defined by $\alpha_*\{z\} = \{\alpha_p z\}$.

Proof In fact the collection of homomorphisms α_p is a chain map (4.36), i.e. writing ∂ for the boundary homomorphism in either K or K_1 , $\partial \circ \alpha_p = \alpha_{p-1} \circ \partial$ for all p . The result follows easily from this (compare 4.37). To prove the assertion about (α_p) it is enough to check that $\partial \circ \alpha_p(\sigma\tau) = \alpha_{p-1} \circ \partial(\sigma\tau)$ whenever $\sigma\tau \in K$, and also $\partial \circ \alpha_p\sigma = \alpha_{p-1} \circ \partial\sigma$. Using the formula for α_p above, 5.2 and the fact that $\partial \circ \partial = 0$ it follows without much difficulty that

$$\partial \circ \alpha_p(\sigma\tau) = \alpha_{p-1} \circ \partial(\sigma\tau) = (\partial\sigma)\tau - (-1)^n v(\partial\sigma)(\partial\tau)$$

where $n = \dim \sigma$. The other assertion follows similarly. $\square$

Next we shall define a homomorphism

$$\beta_p : C_p(K_1) \rightarrow C_p(K).$$

To do this we consider a more general situation.

Definition 5.4. Let K_1 and K be any two simplicial complexes. A *simplicial map*

$$\beta : K_1 \rightarrow K$$

is a map from the set of vertices of K_1 to the set of vertices of K with the property that

$$(w^0 \dots w^p) \in K_1 \Rightarrow \{\beta w^0, \dots, \beta w^p\} \text{ is the set of vertices of a simplex of } K.$$

Note that $\beta w^0, \dots, \beta w^p$ need not be distinct: for example β could map all vertices of K_1 , to the same vertex of K .

Given a simplicial map $\beta : K_1 \rightarrow K$ as above we can define a homomorphism

$$\beta_p : C_p(K_1) \rightarrow C_p(K)$$

by

$$\beta_p(w^0 \dots w^p) = \begin{cases} (\beta w^0 \dots \beta w^p) & \text{if these } p+1 \text{ vertices} \\ & \text{of } K \text{ are distinct} \\ 0 & \text{otherwise} \end{cases}$$

Proposition 5.5. *With the above notation, β_p induces a homomorphism*

$$\beta_* : H_p(K_1) \rightarrow H_p(K)$$

defined by $\beta_\{z\} = \{\beta_p z\}$.*

Proof Again (β_p) is a chain map. The proof will be left to the reader, with the following hint: consider in turn the cases where $\beta w^0, \dots, \beta w^p$ contains $p+1$ distinct vertices of K , has one repetition, and has more than one repetition. In the first case $\beta_{p-1} \circ \partial(w^0 \dots w^p) = \partial \circ \beta_p(w^0 \dots w^p)$ since the formulae for the two sides are identical. In the other cases both sides are zero. $\square$

Returning to the case where K_1 is obtained from K by starring $s = (v^0 \dots v^n)$ at v , there is a simplicial map $\beta : K_1 \rightarrow K$ defined by $\beta v = v^0$ and $\beta w = w$ for $w \neq v$. By 5.5 this gives a homomorphism $\beta_* : H_p(K_1) \rightarrow H_p(K)$.

Theorem 5.6. *With the notation just given, and that of 5.3, α_* and β_* are inverse isomorphisms, so that*

$$\alpha_* : H_p(K) \cong H_p(K_1).$$

Proof It is immediate from the definitions that $\beta_p \circ \alpha_p = \text{identity}$; hence $\beta_* \circ \alpha_*\{z\} = \beta_*\{\alpha_p z\} = \{\beta_p \circ \alpha_p z\} = \{z\}$ for any $\{z\} \in H_p(K)$, i.e. $\beta_* \circ \alpha_* = \text{identity}$.

Next note that, for any $c \in C_p(K_1)$, $\alpha_p \circ \beta_p c - c$ contains only simplexes in $\overline{\text{St}}(v, K_1)$, since β_p does not change anything outside $\overline{\text{St}}(v, K_1)$, and α_p does not change anything outside $\overline{\text{St}}(s, K)$. If $z \in Z_p(K_1)$, it follows that $\alpha_p \circ \beta_p z - z$ is a cycle on $\overline{\text{St}}(v, K_1)$, since α_p and β_p take cycles to cycles. Now $\overline{\text{St}}(v, K_1)$ is the join of $\{v\}$ to the link $\text{Lk}(v, K_1)$ (see 3.29(4)): i.e. the closed star is a cone on the link. Now any p -cycle ($p > 0$) on a cone is homologous to zero. This follows at once from 6.8, or alternatively from the following direct argument.

Let $z' \in Z_p(\overline{\text{St}}(v, K_1))$ and write L for $\text{Lk}(v, K_1)$. Then there are chains $c' \in C_{p-1}(L)$ and $c'' \in C_p L$ such that $z' = vc' + c''$ (thus c' contains the simplexes in z' which have v as a vertex). Hence $0 = \partial z' = c' - v(\partial c') + \partial c''$ from 5.2, and this implies that $c' + \partial c'' = 0$ and $\partial c' = 0$. Hence

$$\partial(vc'') = c'' - v(\partial c'') = c'' + vc' = z'$$

so that $z' \sim 0$.

This shows that $\alpha_p \circ \beta_p z \sim z$, and so $\alpha_* \circ \beta_* = \text{identity}$ for $p > 0$. For $p = 0$, we have $\alpha_0 \circ \beta_0 v - v = v^0 - v \sim 0$ (since $(v^0 v) \in K_1$) and $\alpha_0 \circ \beta_0 w = w$ for $w \neq v$; hence the result, namely $\alpha_* \circ \beta_* = \text{identity}$ for all p . $\square$

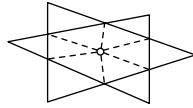
Corollary 5.7 (Invariance under stellar subdivision). *Suppose that K_1 is a stellar subdivision of K , i.e. is obtained from K by a sequence of starrings. Then $H_p(K_1) \cong H_p(K)$ for all p . In particular the isomorphism holds if K_1 is obtained from K by one or more barycentric subdivisions.*

Proof This follows immediately by repeated application of the theorem. $\square$

5.8. The relative case In order to avoid overburdening the above proofs with notation, the absolute case only was considered. Nevertheless the same arguments can be used for a pair (K, L) consisting of a simplicial complex K and a subcomplex L . Thus (K_1, L_1) is a subdivision of (K, L) if K_1 subdivides K and L_1 subdivides L . A stellar subdivision K_1 of K will automatically give a stellar subdivision L_1 of L , and in the case of a single starring the homomorphisms α_* and β_* can be defined as before, but this time between $H_p(K, L)$ and $H_p(K_1, L_1)$. (Note that a simplicial map $\beta : (K_1, L_1) \rightarrow (K, L)$ is a simplicial map $K_1 \rightarrow K$ such that $(w^0 \dots w^p) \in L_1 \Rightarrow \{\beta w^0, \dots, \beta w^p\}$ are the vertices of a simplex of L .) As before, α_* and β_* are inverse isomorphisms, and it follows that $H_p(K_1, L_1) \cong H_p(K, L)$ whenever (K_1, L_1) is obtained from (K, L) by successive starrings.

Remark 5.9. Two simplicial complexes K and L are called *piecewise linearly (p.l.) equivalent* if there are subdivisions K_1 of K and L_1 of L such that K_1 and L_1 are isomorphic. For example, any two closed surfaces with the same standard form are p.l. equivalent (as well as being equivalent in the sense of 2.8). To see this informally superimpose their standard polygons (assumed regular) and subdivide to make a common subdivision of each; the diagram indicates roughly how this is done for two overlapping triangles. The converse is also true: if two surfaces are p.l. equivalent then they have the same standard

form. Compare 2.16(8), which says that Euler characteristic is unaltered by subdivision.

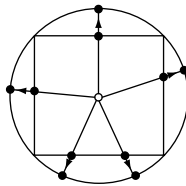


The same sort of subdivision construction as that illustrated above can be used to show that any two triangulations of the same polyhedron, i.e. any two simplicial complexes in $\mathbb{R}^N$ with the same underlying space, have a common subdivision, and hence are p.l. equivalent.

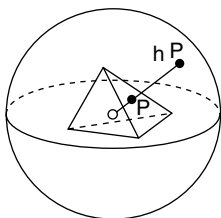
★ Triangulations, simplicial approximation and topological invariance

In this section we shall sketch very briefly the route to a more general invariance theorem than 5.7. Full details appear in many textbooks of algebraic topology, for example the books by Prasolov *Elements of Homology Theory* or Hilton & Wylie listed in the References. There is also a pleasant account of the two-dimensional case in the book of Ahlfors & Sario.

Let X be a subset of a euclidean space $\mathbb{R}^N$. Then certainly X may fail to be the underlying space of any simplicial complex – for example if X is a circle – but the next best thing to hope for is that there is some simplicial complex K such that $|K|$ is homeomorphic to X . Thus taking X to be the unit circle in $\mathbb{R}^2$ and K to be the set of edges and vertices of the square in $\mathbb{R}^2$ with vertices $(\pm \frac{\sqrt{2}}{2}, \pm \frac{\sqrt{2}}{2})$ a homeomorphism $f : |K| \rightarrow X$ can be defined by radial projection from the origin.



Let $X \subset \mathbb{R}^N$. Given a simplicial complex K and a homeomorphism $h : |K| \rightarrow X$ we call K (or, more precisely, h) a *triangulation* of X . In this sense the closed surfaces of Chapter 2 are triangulations of appropriate ‘curved surfaces’ in euclidean space. For example the hollow tetrahedron is a triangulation of a 2-sphere, h being given say by radial projection.



As another example, the unit disk D^2 in $\mathbb{R}^2$ is the set of points (x, y) with $x^2 + y^2 \leq 1$; thus a triangulation of D^2 is a homeomorphism $h : |K| \rightarrow D^2$ where K is a simplicial complex. Now K must be two-dimensional (since h is a homeomorphism); the image of the 1-skeleton of K in D^2 will be a curved graph in the sense of the discussion leading up to Fáry's theorem 1.34. By that theorem there is a homeomorphism $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ taking this curved graph to a straight graph, so that $f \circ h$ defines an isomorphism between K and a simplicial complex in $\mathbb{R}^2$. Thus every triangulation of the disk D^2 can be realized in $\mathbb{R}^2$. With slightly more trouble one can prove, also using Fáry's theorem, that every triangulation of the 2-sphere $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ can be realized in $\mathbb{R}^3$.

When K is a triangulation of X we cannot immediately define the homology groups of X to be those of K , since there could well be another simplicial complex K_1 and a homeomorphism $h_1 : |K_1| \rightarrow X$; we do not know that $H_p(K_1) \cong H_p(K)$ for all p . As an example, K_1 could be any subdivision of K , in which case h_1 could equal h , and then we know the existence of an isomorphism only for stellar subdivisions.

What is needed is a result which asserts the following: Suppose that K_1 and K are oriented simplicial complexes and that $f : |K_1| \rightarrow |K|$ is a homeomorphism. Then $H_p(K_1) \cong H_p(K)$ for all p . Granted this and given h_1 and h as above, we simply take $f = h^{-1} \circ h_1$ to show that the two triangulations of X have isomorphic homology groups. There follows a brief summary of the steps needed to obtain the above result.

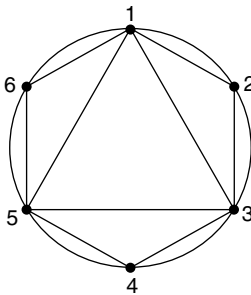
It is worth noting that it is not necessarily true that K_1 and K as above are p.l. equivalent (5.9). At one time it was conjectured that any two triangulations of homeomorphic polyhedra were p.l. equivalent; this was known as the Hauptvermutung (main conjecture). The first counterexample (six-dimensional polyhedra) was found by Milnor (1961), and subsequently it was discovered that there exist 'topological manifolds' of any dimension ≥ 4 which do not admit of any triangulation, and other four-dimensional manifolds which admit of infinitely many distinct p.l. structures. This area is still the subject of research in topology. See Freedman (1982), and also the book edited by Ranicki listed in the References.

Let K_1 and K be any two simplicial complexes and let $f : |K_1| \rightarrow |K|$ be a continuous map. We ‘approximate’ f by a simplicial map $\beta : K_1 \rightarrow K$.

Definition 5.10. A simplicial map $\beta : K_1 \rightarrow K$ is called a *simplicial approximation* to $f : |K_1| \rightarrow |K|$ provided that, for each vertex v of K_1 ,

$$f(\text{St}(v, K_1)) \subset \text{St}(\beta v, K).$$

As an example, the diagram shows a hexagon K_1 (vertices $1, \dots, 6$) and a triangle K (vertices $1, 3, 5$) inscribed in a circle. A continuous map $|K_1| \rightarrow |K|$ is defined by projection towards the centre of the circle, and a simplicial approximation to this is $\beta : K_1 \rightarrow K$ where β takes $1, 2, 3, 4, 5, 6$ to, respectively, $1, 3, 3, 5, 5, 1$. On the other hand consider the inverse map $g : |K| \rightarrow |K_1|$ given by outward projection. There is no simplicial map $K \rightarrow K_1$ which is a simplicial approximation to g , since $g(\text{St}(1, K))$ is not contained in the star of any vertex of K_1 . However K can be barycentrically subdivided – once is enough, giving K' – so that the map $g : |K'| \rightarrow |K_1|$ does have a simplicial approximation. In general, the following result holds.



5.11. Simplicial approximation theorem Given simplicial complexes K_1 and K and a continuous map $f : |K_1| \rightarrow |K|$ there exists an integer r and a simplicial map $\beta : K_1^{(r)} \rightarrow K$ (the superscript denoting r th barycentric subdivision) which is a simplicial approximation to f . $\square$

Suppose, then, that K_1, K and f are as in 5.11; by 5.7 there is an isomorphism $\gamma_* : H_p(K_1) \rightarrow H_p(K_1^{(r)})$ and so a homomorphism

$$f_* : H_p(K_1) \xrightarrow{\gamma_*} H_p(K_1^{(r)}) \xrightarrow{\beta_*} H_p(K)$$

for each p . One vital, and far from obvious, fact about f_* is that it does not depend on r or β , so long as β is a simplicial approximation to f . Thus it

really depends only on f , and the notation f_* is justified. Another important fact is this:

Lemma 5.12. *Let $g : |K_2| \rightarrow |K_1|$ and $f : |K_1| \rightarrow |K|$ be continuous maps, where K_2, K_1 and K are (oriented) simplicial complexes. Then $(f \circ g)_* = f_* \circ g_* : H_p(K_2) \rightarrow H_p(K)$ for each p .* $\square$

The result required now follows at once by taking $K_2 = K$ and f a homeomorphism with inverse g . Hence:

5.13. Topological invariance of homology groups *Let K_1 and K be (oriented) simplicial complexes and $f : |K_1| \rightarrow |K|$ be a homeomorphism. Then $f_* : H_p(K_1) \rightarrow H_p(K)$ is an isomorphism for each p .* $\square$

★

★ Appendix on barycentric subdivision

Let s be an n -simplex, $s = (v^0 \dots v^n)$. The *barycentre* of s is the point $\widehat{s} = \frac{1}{n+1}(v^0 + v^1 + \dots + v^n)$. Thus the barycentre of a 0-simplex (v) is v , the barycentre of a 1-simplex (v^0v^1) is its mid-point, and the barycentre of a 2-simplex ($v^0v^1v^2$) is its centroid.

The *first barycentric subdivision* K' of a simplicial complex K is the simplicial complex with vertex set $\{\widehat{s} : s \in K\}$, and with simplexes $t = (\widehat{s}^0 \widehat{s}^1 \dots \widehat{s}^q)$ where $s^0 < s^1 < \dots < s^q$ and $s^0, \dots, s^q$ are simplexes of K . (The superscript is just for indexing, and does not necessarily give the dimension of s^i .) Notice that $t \subset s^q$; to see that K' is a subdivision of K we must therefore show that $|K| \subset |K'|$. If $x \in |K|$ then $x \in s$ for some $s \in K$ – we can take the s of lowest dimension for which this is true, so that if $s = (v^0 \dots v^p)$ then $x = \lambda_0 v^0 + \dots + \lambda_p v^p$ where the barycentric coordinates are all > 0 . We can also assume $\lambda_0 \geq \lambda_1 \geq \dots \geq \lambda_p$. Let $s^i = (v^0 \dots v^i)$ ($i = 0, \dots, p$); then it is straightforward to check that $x \in (\widehat{s}^0 \widehat{s}^1 \dots \widehat{s}^p)$: in fact

$$\begin{aligned} x &= (\lambda_0 - \lambda_1) \widehat{s}^0 + 2(\lambda_1 - \lambda_2) \widehat{s}^1 \\ &\quad + \dots + p(\lambda_{p-1} - \lambda_p) \widehat{s}^{p-1} + (p+1) \lambda_p \widehat{s}^p. \end{aligned}$$

If L is a subcomplex of K then L is automatically subdivided into L' when K is subdivided into K' . The reader may like to check the following assertions.

- (1) $\widehat{s}$ is a vertex of L' if and only if $s \in L$.
- (2) If t is a simplex of K' such that all vertices of t belong to L' , then $t \in L'$.

- (3) Suppose that $(vw) \in K'$, where v is a vertex of L' and w is not a vertex of L' ; then w is not a vertex of K (hence is the barycentre of a simplex of K of dimension > 0).
- (4) Suppose that $s \in K$ and that no vertex of s belongs to L . Then there is no edge $(vw) \in K'$ with $v \in L'$ and w the barycentre of a face of s . ★